Behavioral Latency as Weak Event-Time Supervision for EEG Reaction-Time Decoding

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Abstract

Single-trial EEG analyses are often organized around events and latencies, such as stimulus onset, response execution, ERP component timing, and movement onset. Yet EEG-based reaction-time prediction is commonly formulated as scalar regression from

a fixed stimulus-locked window, treating reaction time as a label attached to the whole segment rather than as behavioral evidence about when a response-relevant event occurs within post-stimulus dynamics. Here we reformulate trial-wise reaction-time decoding as event-time posterior modeling. Instead of predicting a single RT directly, the model estimates a posterior distribution over response-relevant event times, $p(t_{\text{event}} \mid X)$, and reads out RT as the posterior expectation. This converts a scalar behavioral latency label into weak event-time supervision over the post-stimulus EEG window.

We evaluate this formulation on the Healthy Brain Network contrast change detection EEG task under a subject-disjoint, release-separated protocol. Architecture and readout controls show that the improvement is not explained by architecture scale or by a differentiable expectation readout alone. Distributional event-time supervision yields lower held-out error than the strongest scalar and temporal-readout baselines. Loss ablations and five-seed checks support the interpretation that the gain comes from supervising the event-time distribution, not merely from using a soft-argmax readout. EventNLL provides a probabilistic latent-event alternative, treating observed RT as a noisy realization of an unobserved response-relevant event time.

Beyond point prediction, the event-time posterior makes temporally resolved evaluation possible rather than limiting assessment to scalar nRMSE. Posterior width, target-aligned mass, mode-mean disagreement, and interval coverage characterize whether predictions are localized, diffuse, overconfident, or miscalibrated. We further introduce shifted-crop inference as a shortcut-vs-localization diagnostic: under temporal crop shifts, event-time models move in a more localizer-like direction, while also showing that fixed-window training does not fully solve shift-equivariant event localization. Together, these results support event-time distribution modeling as a probabilistic and interpretable formulation for linking single-trial EEG dynamics to behavioral timing. The same event-time view is naturally relevant to EEG settings in which the target has a latency interpretation, including ERP latency estimation, movement onset decoding, and other latency-varying neural or behavioral phenomena.

1 Introduction

Behavioral timing provides a direct link between neural dynamics and behavior. In a reaction-time task, the observed response latency is not only a behavioral score; it is the observable outcome of sensory encoding, decision formation, motor preparation, and response execution unfolding over hundreds of milliseconds. This makes reaction time (RT) a natural test case for asking how supervised EEG models should represent temporally structured behavioral evidence.

Many EEG analyses already exploit temporal structure through event alignment, response locking, and latency variability. Across stimulus-locked ERPs, response-locked error signals, and movement-related potentials, timing is often part of the scientific object being studied, not just a nuisance to be averaged away. Trial-to-trial latency variability can carry behaviorally meaningful information and may be obscured both by averaging and by supervised models that collapse temporal structure into window-level labels (Jung et al., 1998; Ouyang et al., 2017).

Supervised EEG decoding often uses this temporal signal in a less explicit way. A fixed stimulus-locked EEG window is mapped to a class label or scalar behavioral variable, even when the target has a natural time-of-occurrence interpretation. In RT decoding, this means that a delayed behavioral event is treated as a property of the whole EEG segment. A scalar predictor may therefore achieve low error by learning subject-level slowness, trial difficulty, or stimulus-locked response tendencies, without representing when the response-relevant event occurs within the window.

We address this issue by treating behavioral RT as weak event-time supervision. The supervision is weak because the button press is not a direct annotation of neural event onset; it is a noisy behavioral observation produced by sensory, decision, and motor processes. Nevertheless, it constrains the timing of response-relevant dynamics. Given an EEG window X, the model estimates a posterior distribution $p(t_{\text{event}} \mid X)$ over response-relevant event times and derives the scalar RT prediction as the posterior expectation, preserving compatibility with standard scalar metrics such as nRMSE while making the timing semantics of RT explicit.

This formulation is complementary to recent work on stronger EEG architectures, self-supervised pretraining, and foundation-style EEG models, which aim to improve representation learning across subjects, tasks, and recording setups (Wang et al., 2024; Jiang et al., 2024; Chen et al., 2024; Wang et al., 2025; Ouahidi et al., 2025; Yue et al., 2025). We focus on a different question: when the target is a behavioral latency, does the representation of the target also matter, not only the representation capacity of the backbone? For latency-structured EEG targets, the bottleneck may lie not only in architecture scale or pretraining, but also in whether the learning objective matches the temporal semantics of the label.

We study this question using the Healthy Brain Network contrast change detection EEG task (Shirazi et al., 2024), following the reaction-time prediction setting introduced by the EEG Foundation Challenge (Aristimunha et al., 2025; EEG Challenge, 2025). Each trial provides a fixed 2 s stimulus-locked EEG window, and the goal is to predict the participant's RT. We evaluate models under a subject-disjoint, release-separated protocol, using earlier HBN releases for training and development and a held-out labeled release for final reporting.

This distinction determines the control structure of our experiments. We compare direct scalar regression, temporal-readout regression, time-only segmentation, and distributionally supervised event-time models to separate architecture capacity, differentiable expectation readout, scalar timing loss with a temporal head, and full posterior supervision. The central test is whether supervising the event-time distribution provides a better inductive bias than treating RT as an ordinary scalar label.

The posterior is also scientifically useful because it changes what can be evaluated from each trial. Posterior width, target-aligned mass, mode-mean disagreement, and interval coverage characterize localization, uncertainty, and calibration, distinguishing models with similar scalar error. We further introduce shifted-crop inference as a shortcut-vs-localization diagnostic: a crop-relative event localizer should shift its predicted crop-relative event time opposite to the crop shift, whereas a crop-invariant scalar shortcut should remain closer to fixed absolute timing.

Our results support this formulation-first view. Compact task-specific scalar models provide strong baselines, and larger supervised or foundation-style architectures trained from scratch do not by themselves close the gap. Distributional event-time supervision improves held-out prediction over scalar and temporal-readout controls and remains stable in five-seed checks, while time-only and readout ablations show that the gain is not explained by soft-argmax alone. Posterior-geometry and shifted-crop analyses further reveal temporal evidence and partial localization behavior hidden by scalar metrics.

Although our experiments focus on RT prediction, the broader claim is that neural or behavioral targets with time-of-occurrence semantics should not automatically be reduced to static scalar regression. Event-time posteriors may also be useful for ERP latency estimation, movement onset decoding, error-related responses, and other latency-varying EEG phenomena.

Our contributions are:

1. We recast behavioral RT labels as weak event-time annotations and formulate EEG RT decoding as event-time posterior modeling rather than direct scalar regression.

2. We evaluate this formulation under a subject-disjoint, release-separated HBN-EEG protocol against compact scalar regression, temporal-readout regression, time-only segmentation, and standardized external EEG architecture controls.

3. We show through loss ablations and five-seed robustness checks that distributional event-time supervision improves over scalar, temporal-readout, and time-only controls, and we evaluate EventNLL as a latent event-time likelihood in which observed RT is treated as a noisy realization of an unobserved response-relevant event.

4. We use posterior geometry and shifted-crop inference to separate scalar accuracy from temporal evidence, uncertainty, shortcut behavior, and partial event localization.

2 Related Work

2.1 Event-Centered EEG Analysis and Latency Variability

EEG is often analyzed through events and latencies. Stimulus onset, response execution, error-related activity, movement onset, ERP component timing, and other temporally localized phenomena define not only what occurs in a trial, but also when task-relevant neural or behavioral processes unfold. This event-centered view is reflected in annotation practice: for example, Hierarchical Event Descriptors were introduced to describe laboratory and real-world EEG events in a machine-readable way for large-scale EEG analysis (Bigdely-Shamlo et al., 2016). It is also central to single-trial ERP analysis, where averaging can obscure latency variability across trials. ERP-image visualization and related single-trial methods show that sorting trials by reaction time can reveal performance-linked temporal structure (Jung et al., 1998), while RIDE and subsequent reviews emphasize that intra-subject ERP latency variability can be estimated and can carry behaviorally meaningful information (Ouyang et al., 2011, 2017).

Response- and movement-related EEG provide additional examples of event-centered analysis. Lateralized readiness potentials have been used to study response preparation and response execution in time-resolved EEG (Coles, 1989), and error-related negativities provide response-locked signatures of error commission and performance monitoring (Gehring et al., 1993; Holroyd and Coles, 2002). Movement-related cortical potentials, including the Bereitschaftspotential, further show that EEG dynamics can be organized around movement onset and premovement preparation (Kornhuber and Deecke, 1965; Shibasaki and Hallett, 2006). This body of work motivates the broader premise that latency and event timing are not incidental details of EEG analysis; they are often part of the scientific object being studied.

This perspective also connects to decision-time accounts of behavior. Diffusion and accumulation-to-bound models treat RT variability as a consequence of latent temporal dynamics in evidence accumulation and decision formation (Ratcliff and McKoon, 2008; O'Connell et al., 2012; Kelly and O'Connell, 2013). We do not fit a cognitive process model in this work. Instead, we use the weaker and more task-agnostic premise that an observed RT is a noisy behavioral latency associated with an unobserved response-relevant event time.

Explicit EEG event detection is also well established in settings where annotated onsets, offsets, or intervals are available. Sleep-event detectors, for example, estimate when transient events such as spindles and K-complexes occur rather than predicting only a window-level label (Chambon et al., 2019; Tapia-Rivas et al., 2024). These settings show that event detection is natural for EEG. However, they typically rely on explicit event annotations, such as expert-marked onsets, offsets, intervals, or external sensors. Our setting is different: we do not begin with manually annotated neural event times. We begin with a behavioral latency, RT, which is recorded as a scalar but implicitly provides a weak time-of-occurrence label. Thus, the behavioral response provides timing evidence, but not a manually annotated neural event onset.

2.2 Fixed-Window EEG Decoding of Latency-Structured Targets

Modern supervised EEG decoding often maps a fixed stimulus-locked or task-locked EEG window to a scalar or class label. This formulation is convenient and aligns with standard benchmark metrics, but it can collapse temporal structure when the target itself has a time-of-occurrence interpretation. Reaction time is a clear example: it is evaluated as one scalar per trial, but it denotes the latency of a behavioral response following stimulus onset.

EEG-based RT estimation has commonly been treated in this scalar-prediction form. Prior work has estimated RT from EEG using regression-oriented feature pipelines, including Riemannian geometry features for BCI regression (Wu et al., 2017), as well as deep single-trial EEG models trained to predict visual-stimulus reaction time directly

(Chowdhury et al., 2020). The EEG Foundation Challenge provides a standardized scalar benchmark for trial-wise RT prediction, using normalized RMSE as the evaluation metric (Aristimunha et al., 2025; EEG Challenge, 2025). We keep this scalar evaluation protocol. Our question is complementary: because RT is a response latency inside the post-stimulus EEG window, can the same label be used internally as weak event-time supervision? In this view, the challenge task is not a limitation but a useful testbed for comparing scalar regression, temporal-readout regression, and event-time posterior modeling under the same metric.

This distinction matters for model interpretation. A scalar regressor may learn trial difficulty, subject-level response tendency, or stimulus-locked temporal priors without representing where the response-relevant event occurs inside the input window. A temporal readout alone does not fully resolve this issue: a model can predict weights over time bins and use a soft-argmax expectation while still being trained only through scalar RT error. Our work separates these possibilities by comparing direct scalar regression, temporal-readout regression, time-only segmentation, and distributionally supervised event-time models.

2.3 Distributional and Time-to-Event Output Modeling

From a machine-learning perspective, latency-structured EEG targets motivate output distributions rather than only point estimates. Label distribution learning replaces a single hard label with a distribution over related labels, which is useful when neighboring labels share metric structure or when target ambiguity is meaningful (Geng, 2016). For RT, neighboring labels are ordered time bins rather than unrelated classes, making a smoothed event-time distribution a natural supervision target.

Time-to-event modeling provides a complementary probabilistic language for output distributions over ordered event times. Classical survival analysis and modern neural time-to-event models estimate distributions over event times rather than only scalar predictions or risk scores (Cox, 1972; Lee et al., 2018). Distributional evaluation also makes calibration and uncertainty part of the modeling problem: posterior width, coverage, likelihood scores, and proper scoring rules can distinguish models with similar point errors but different uncertainty behavior (Chapfuwa et al., 2023; Haider et al., 2020).

We use this literature as an output-distribution language for finite EEG windows rather than as a standard survival-analysis setting. We do not study long-horizon risk, censoring, or competing clinical outcomes. Each trial provides a finite post-stimulus EEG window and an observed behavioral latency. We therefore use the time-to-event view locally: the model predicts a posterior over response-relevant event times within the represented window, and the scalar RT prediction is obtained as the posterior expectation. EventNLL gives a probabilistic version of this formulation by treating observed RT as a noisy realization of a latent response-relevant event time.

2.4 Relation to EEG Backbones, Pretraining, and Robust Decoding

A separate line of EEG decoding work addresses cross-subject, cross-session, and cross-dataset variability through stronger input representations. Classical approaches include Riemannian covariance-based decoding and alignment methods for reducing subject or session mismatch (Barachant et al., 2012; Jayaram et al., 2016; Zanini et al., 2018; He and Wu, 2020). Deep learning approaches pursue related goals through domain adaptation, moment matching, supervised EEG architectures, self-supervised pretraining, and foundation-style EEG backbones (Ganin et al., 2016; Sun and Saenko, 2016; Kostas et al., 2021; Wang et al., 2024; Jiang et al., 2024; Chen et al., 2024; Wang et al., 2025; Ouahidi et al., 2025; Yue et al., 2025).

Our contribution is complementary to these input-representation approaches. We do not propose a new general-purpose EEG backbone, invariant representation loss, or pretraining strategy. Instead, we focus on the output representation and supervision. When the target is a latency, the label is not only a scalar value to regress; it is also a weak annotation of when a relevant event occurred. In our experiments, external EEG architectures and foundation-style models trained from scratch serve as controlled architecture baselines. The main question is whether aligning the learning objective with the event-time semantics of RT improves decoding and yields more interpretable trial-wise temporal evidence.

Taken together, prior work provides strong traditions for event-centered EEG analysis, fixed-window supervised decoding, explicit EEG event detection, and time-to-event modeling. What remains missing is a controlled EEG RT formulation that connects scalar benchmark evaluation, temporal-readout controls, distributional event-time supervision, and posterior-level diagnostics. We fill this gap by treating behavioral latency as weak event-time supervision, learning a posterior over response-relevant latencies, and evaluating that posterior as temporal evidence rather than only as a source of a scalar RT estimate.

3 Dataset and Evaluation Protocol

3.1 HBN-EEG Dataset and Reaction Time Task

We use the reaction-time prediction task introduced by the NeurIPS EEG Foundation Challenge (EEG Challenge, 2025; Aristimunha et al., 2025), based on curated releases of the HBN-EEG dataset (Shirazi et al., 2024). Our focus is trial-wise reaction-time prediction from the contrast change detection (CCD) paradigm. The challenge provides the standardized task definition; our main evaluation uses labeled HBN releases rather than the original hidden leaderboard. The broader HBN-EEG resource comprises recordings from more than 3,000 participants aged 5–21 collected across a standardized battery of six EEG tasks spanning both passive and active paradigms (Shirazi et al., 2024; EEG Challenge, 2025).

HBN-EEG is distributed in a Brain Imaging Data Structure (BIDS)-compatible format with detailed event annotations using Hierarchical Event Descriptors (HED),

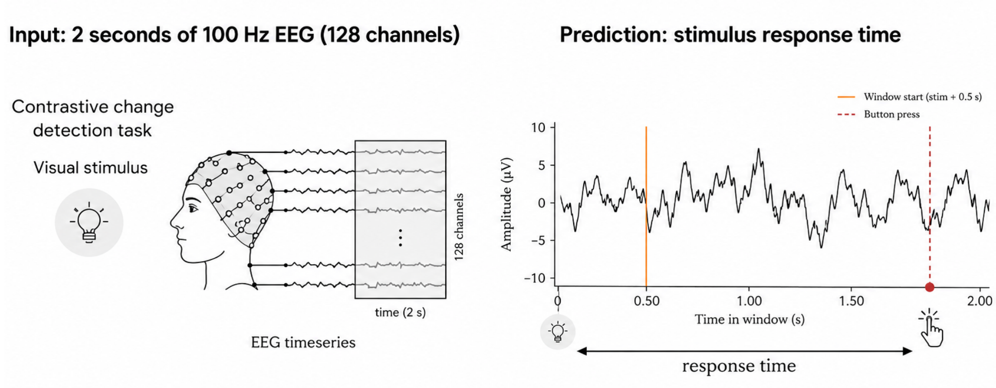

Figure 1: Reaction-time prediction as event-time modeling. Each trial provides a 2 s stimulus-locked EEG window from the CCD paradigm (0.5–2.5 s after stimulus onset; 128 EEG channels at 100 Hz). The observed RT is represented as a smoothed distribution over post-stimulus time bins. The model predicts an event-time distribution, which is converted to a scalar RT prediction using a temperature-scaled soft-argmax readout.

enabling reproducible preprocessing and time-locked analyses (Shirazi et al., 2024). In the original acquisition, EEG was recorded in a sound-shielded environment using a high-density 128-channel EGI Geodesic HydroCel system at 500 Hz, referenced at Cz, with an online 0.1–100 Hz bandpass and electrode impedances maintained below 40 kΩ. For the benchmark derivative, recordings are band-pass filtered (0.5–50 Hz) and downsampled to 100 Hz to standardize modeling across releases (EEG Challenge, 2025). The released derivatives include 129 channels (128 EEG channels plus a reference); throughout this work we train models on the 128 EEG channels and exclude the reference channel from the learnable input.

The CCD task is the active HBN-EEG paradigm used in this study. In each trial, participants viewed two concentric, flickering grating stimuli, one left-tilted and one right-tilted. After a variable delay, the contrast of one stimulus increased, making it perceptually dominant. Participants were instructed to respond as quickly as possible to indicate whether the dominant grating was left- or right-tilted, after which they received immediate accuracy feedback.

Each model input is a fixed-length 2 s EEG segment, represented as 128 EEG channels sampled at 100 Hz and time-locked to the stimulus event. The fixed inference window corresponds to 0.5–2.5 s after stimulus onset (Figure 1). The released target is a scalar RT for each trial, and scalar performance is evaluated using normalized RMSE (nRMSE), defined as RMSE divided by the standard deviation of the true targets (EEG Challenge, 2025). Because the label reflects when a response occurs after stimulus onset, this task naturally motivates modeling RT as an event-time quantity rather than as a static scalar regression target.

3.2 Preprocessing and Evaluation Protocol

We evaluate models using a fixed split over HBN releases. Releases R1–R8 are used for fitting the base EEG models. Releases R9–R10 form the development partition used for checkpointing, calibration, and protocol diagnostics. When optional stacking is reported, meta-model inputs on R9–R10 are generated using subject-disjoint out-of-fold predictions.

Release R11 serves as the labeled test release for final reporting. Model development, calibration, and selection are completed before R11 scoring. Main performance results are reported on R11 with subject-level bootstrap confidence intervals. The original hidden CodaBench evaluation from the EEG Foundation Challenge is reported separately in the Appendix as benchmark context.

For posterior-geometry analyses, we distinguish scalar RT evaluation from event-time support diagnostics. Scalar nRMSE is computed on all R11 trials, whereas posterior-shape summaries are restricted to trials whose observed RT lies inside the modeled event-time grid spanning 0.50–2.49 s from stimulus onset.

All neural models receive the same input normalization. For each 2 s trial window and each EEG channel, we subtract that channel's temporal mean within the window and divide by its temporal standard deviation within the same window. This transformation is computed independently within each input window and is applied identically across models.

Unless otherwise stated, training uses a fixed mild augmentation recipe. Channel dropout is applied with probability 0.25 and removes at most 30% of channels. Temporal cutout is applied with probability 0.25 and masks a contiguous 10–50 sample interval, corresponding to at most 0.5 s in a 2 s, 100 Hz window. Gaussian noise is applied with probability 0.3 with standard deviation sampled as $0.01 + U(0, 0.01)$. Mixup and stronger corruption recipes are excluded from the default protocol and treated only as appendix diagnostics.

4 Methods

4.1 Scalar Regression Baselines

We begin with baselines that treat the task as direct reaction-time regression: a stimulus-locked 2 s EEG crop $X \in \mathbb{R}^{C\times T}$ with $T = 200$ samples (100 Hz) is mapped to a scalar RT.

Consistent with the low-SNR and distribution shift setting, we focus on lightweight backbones that are less prone to overfitting and enable extensive ablations, calibration, and ensembling.

All direct-regression baselines use the fixed 2 s input window, the per-window standardization described above, no mixup, batch size 128, patience 20, and the same release-separated train/development/holdout protocol. The default augmentation is the mild channel-dropout, temporal-cutout, and Gaussian-noise recipe described in the dataset protocol; stronger corruptions are retained only as appendix diagnostics.

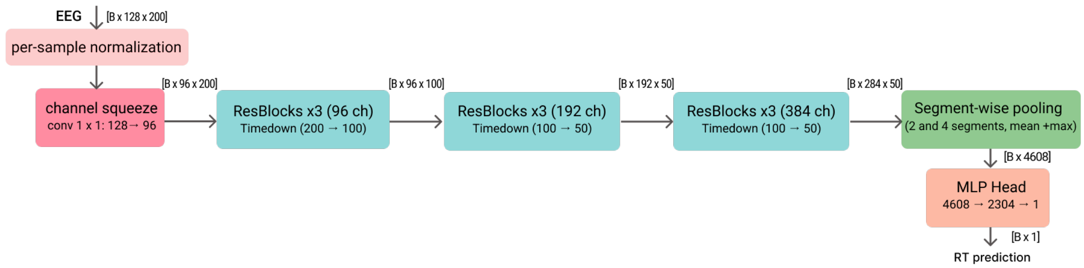

Figure 2: MSP-CNN direct RT regression baseline architecture. A lightweight 1-D temporal ConvNet maps a stimulus-locked 2 s EEG crop (128 channels, 100 Hz) to a scalar reaction-time prediction. Per-sample normalization and a learned 1×1 channel-squeeze layer (128→96) are followed by residual depthwise-separable stages with anti-aliased temporal downsampling, segment-wise pooling (2 and 4 segments; mean+max), and an MLP head.

For clarity, we use mechanism-level names in the manuscript. MSP-CNN denotes a multiscale segment-pooling CNN. It maps the full EEG window to scalar RT through temporal segment statistics and an MLP head. These models serve as reference points for isolating the effect of making temporal structure explicit in Section 4.3. In contrast to the segmentation formulation, direct scalar regression treats the window as a static input-output regression problem, without an explicit representation of response timing.

The main MSP-CNN regression backbone (Figure 2) is a lightweight 1-D temporal ConvNet with: (i) per-sample normalization to reduce cross-subject amplitude variation, (ii) a learned channel-squeeze layer (1×1 conv, 128→96) that forms data-driven electrode mixtures, and (iii) an EfficientNet-like temporal hierarchy with widening stages of residual depthwise-separable blocks (96→192→384 channels) and anti-aliased temporal downsampling. Coarse timing cues are retained using segment-wise statistical pooling: the time axis is split into 2 and 4 segments, mean and max are computed per segment, and the resulting features are passed to a small MLP head.

To contextualize these task-specific regressors against established EEG backbones, we also evaluate external architectures adapted to scalar RT regression. These include classical convolutional EEG baselines, Deep4Net and ShallowFBCSPNet (Schirrmeister et al., 2017), the compact EEGNet architecture (Lawhern et al., 2018), TIDNet's thinker-invariant convolutional design (Kostas and Rudzicz, 2020), the convolutional-transformer EEGConformer (Song et al., 2023), the attention temporal convolutional ATCNet architecture (Altaheri et al., 2022), and the LaBraM foundation-style architecture (Jiang et al., 2024). All external architectures are evaluated as supervised models trained from scratch under the same fixed preprocessing protocol. Pretrained weights are not used. The main tables report the wrapped/standardized variants because they receive the same per-window standardization as our models; unwrapped external runs are treated as sanity checks in the Appendix.

Release-separated baseline comparison. Table 1 reports direct-regression and external-backbone baselines under the release-separated protocol, repeated over five

Table 1: Direct-regression and wrapped external backbone baselines under the release-separated protocol. Values are mean ± sample standard deviation over seeds 2025–2029. Lower nRMSE is better.

Model	Role	Valid nRMSE ↓	R11 nRMSE ↓
MSP-CNN	compact scalar regression baseline	0.900595 ± 0.005096	0.899795 ± 0.008045
ETR-CNN	temporal-readout direct regression	0.900822 ± 0.006043	0.897679 ± 0.006773
ETR-CNN large	capacity ablation	0.897233 ± 0.004022	0.892783 ± 0.004161
TIDNet wrapped	strongest external supervised baseline in this comparison	0.923506 ± 0.002357	0.919204 ± 0.002739
EEGConformer wrapped	modern supervised transformer-style baseline	0.918845 ± 0.002634	0.928656 ± 0.005658
EEGNet wrapped	canonical compact EEG baseline	0.935037 ± 0.005384	0.933504 ± 0.002832
LaBraM wrapped	foundation-style architecture from scratch	0.930388 ± 0.006065	0.932722 ± 0.008568
Deep4Net wrapped	classical convolutional baseline	0.926883 ± 0.004521	0.925972 ± 0.004354
ShallowFBCSPNet wrapped	classical filter-bank spatial convolution baseline	0.932353 ± 0.001632	0.934285 ± 0.002423
ATCNet wrapped	supervised conv/attention/TCN baseline	0.968640 ± 0.018283	0.966578 ± 0.015093

seeds. The ETR-CNN large capacity variant is the strongest direct-regression baseline under the fixed protocol, while the compact MSP-CNN and base ETR-CNN models remain competitive. Wrapped external architectures are substantially better controlled than their unwrapped sanity-check counterparts, but remain behind the task-specific regression baselines on R11.

Despite strong performance and fast training, direct regression does not explicitly represent the event-timing structure of the task. Under heavier temporal jitter and cropping, optimization tended to over-focus on local waveform micro-structure rather than learning *when* the response-related event occurs. This limitation motivated the segmentation reformulation in Section 4.3, which makes response timing explicit via an event-time distribution.

4.2 Temporal-Readout Regression

Temporal-readout regression is an intermediate baseline between scalar pooling and full event-time supervision. These models predict a distribution over time bins and read RT as an expectation, but the supervision remains scalar RT or an RT-derived readout loss rather than a full event-time posterior objective. Table 2 summarizes representative regression-style architecture families; the release-separated performance comparison is reported in Table 1.

All baselines implement the same high-level pipeline: *(a)* per-sample temporal standardization, *(b)* a learnable 1×1 channel-mixing projection ($C \to c_0$), and *(c)* a lightweight depthwise-separable residual convolutional backbone inspired by DS-ResNet style and MobileNet/Xception (Howard et al., 2017; Chollet, 2017). They differ mostly in how the final temporal feature map is converted into an RT estimate, that is, the *temporal readout*. In all cases, supervision remains scalar RT (or an RT-derived target) without explicit event-time labels; the temporal soft-argmax head is used only as a readout parameterization, not as event-time supervision.

We consider three temporal readout families. **Segment-stat pooling with an MLP** (MSP-CNN family) applies segmented mean and max pooling over time to encode

Table 2: Regression-style architecture families used before the event-time segmentation pivot. *DS* = depthwise-separable. Current release-separated results are reported in Table 1.

Model	Key design (high-level)	Temporal readout	Manuscript role	Size (MB)
MSP-CNN-S	**Compact DS-residual encoder** Shallow depth and small kernels emphasize local temporal motifs. Strong temporal compression yields coarse features suited to pooling.	Segment-stat pooling + MLP	compact MSP-CNN baseline	4.95
MSP-CNN-L	**Higher-capacity DS-residual encoder** Wider/deeper stack and larger kernels increase temporal context and channel capacity. Same coarse temporal hierarchy, but with stronger representational power.	Segment-stat pooling + MLP	capacity ablation	45.66
ETR-CNN-Dilated	**Dilated DS-residual backbone** Dilations expand receptive field while delaying temporal compression relative to MSP-CNN. Designed to support distributional timing rather than coarse pooling.	Temporal soft-argmax	ETR-CNN baseline	49.24
Recurrent ETR-CNN (ACT)	**Tied-weight recursive DS core** Iterative refinement with step conditioning; adaptive computation via ACT halting. Large effective depth at low parameter cost, paired with distributional timing.	ACT temporal soft-argmax	exploratory readout variant	1.67

coarse early and late cues (conceptually similar to segment-based aggregation in TSN (Wang et al., 2016)), followed by an MLP regressor. **Temporal soft-argmax** (ETR-CNN) predicts a distribution over time bins and reads RT as its expectation via a soft-argmax (coordinate-regression style) readout (Nibali et al., 2018). **ACT temporal soft-argmax** (recurrent ETR-CNN) refines the time distribution iteratively using tied weights with adaptive computation time (ACT) style halting (Graves, 2016) and FiLM conditioning (Perez et al., 2018), producing a halting-weighted mixture distribution before the same expectation readout. ETR-CNN additionally uses dilated convolutions to expand receptive field while delaying temporal compression relative to MSP-CNN (Yu and Koltun, 2016).

Temporal-readout diagnostic. The release-separated comparison in Table 1 shows that temporal-readout regression improves over scalar segment pooling: ETR-CNN reaches 0.897679 ± 0.006773 R11 nRMSE versus 0.899795 ± 0.008045 for MSP-CNN, and the larger ETR-CNN variant improves further to 0.892783 ± 0.004161. This indicates that temporal capacity and expectation-style readout are useful direct-regression

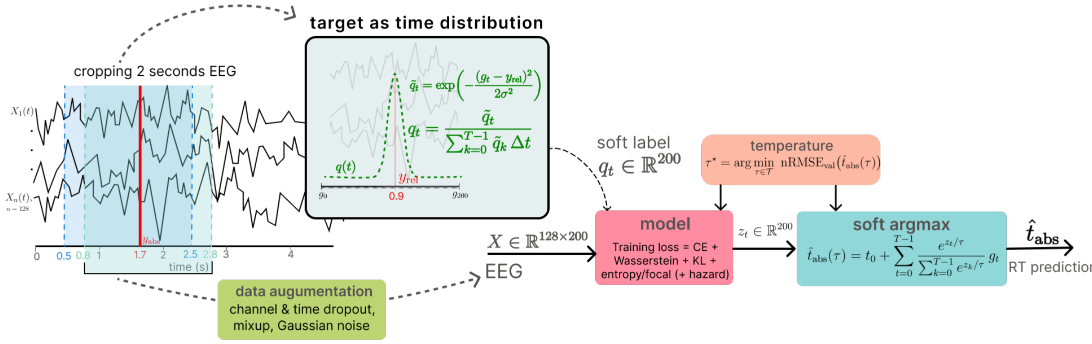

Figure 3: Temporal segmentation pipeline for RT prediction. For each trial we cache a longer stimulus-locked segment (4 s in our implementation) and sample a 2 s subwindow with start-time jitter for training; the RT label is shifted into the sampled window and a Gaussian soft target q_t is formed. Per-time logits z_t are produced by a 1-D segmentation model and are optimized with distribution-matching losses (e.g., soft-label CE, CDF-based Wasserstein-1, KL, and mild entropy penalty) or EventNLL objectives. The main readout is a temperature-scaled softmax posterior; hazard/survival readouts and alternate observation kernels are evaluated as compact extensions. RT is obtained by temperature-scaled soft-argmax, with $\tau^\star$ selected to minimize validation nRMSE.

controls. However, the model is still trained through scalar RT error rather than to match the geometry of an event-time posterior. These considerations motivate the segmentation formulation below, which makes response timing explicit via event-time supervision and distribution-matching losses.

4.3 Event-Time Segmentation Framework

To make event timing explicit and to enable strong crop augmentation, RT prediction was reformulated as *temporal segmentation*: instead of regressing a scalar directly, we predict a distribution over time bins and read out RT as its expectation (Figure 3). Compared to distributional readouts used only as a final head, the key difference here is that supervision is defined in time coordinates and shifted consistently under crop jitter, so augmentation preserves the event-time target by construction. In practice, this produced a large performance gain and faster, more stable convergence, while making crop augmentation safer because the supervision explicitly encodes *where* the event occurs. This is our RT instantiation of explicit temporal structure: behavior is modeled through event timing rather than static regression.

Notation and crop sampling. Let a 2 s crop be $X \in \mathbb{R}^{C\times T}$ with sfreq $= 100$ Hz, $T = 200$ and $\Delta t = 1/\text{sfreq}$. For training-time augmentation, we retain a longer stimulus-locked segment of length 4 s (0–4 s post-stimulus) and sample a 2 s subwindow from it, introducing temporal jitter while preserving trial identity. The crop start is clipped so that the labeled response remains inside the sampled 2 s window (label-preserving

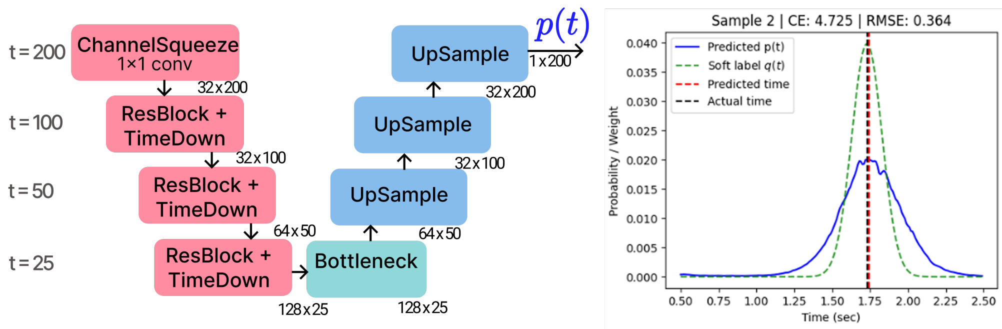

Figure 4: ETS-U-Net for RT segmentation. A lightweight 1-D U-Net maps a 2 s, 128-channel EEG crop to per-time logits and an event-time distribution $p(t)$. The encoder applies a 1×1 channel-squeeze layer followed by residual blocks with temporal downsampling (`TimeDown`); a bottleneck aggregates context, and the decoder upsamples with skip connections to recover time-resolved outputs. The right panel shows an example prediction $p(t)$ and Gaussian soft target $q(t)$, together with the resulting predicted vs. true response time.

jitter). During validation and official inference, we always use the fixed stimulus-locked 2 s window with offset $t_0 = 0.5$ s (0.5–2.5 s), so that $t_{\text{abs}} = t_0 + t_{\text{rel}}$.

ETS-U-Net Architecture. The ETS-U-Net family is used to map a stimulus-locked EEG crop $X \in \mathbb{R}^{C \times T}$ to per-time logits $z_{0:T-1}$ (Figure 4). All segmentation variants follow a 1-D U-Net style encoder-decoder with skip connections for precise localization (Ronneberger et al., 2015). The encoder progressively downsamples the temporal axis to build multi-scale context, while the decoder upsamples and fuses encoder features to recover per-time outputs at the original 200-sample resolution (internal down/up-sampling is used only to form multi-scale features). Each scale is refined with lightweight residual blocks (ResNet-style skips) (He et al., 2016) built from depthwise-separable 1-D convolutions for computational efficiency (Howard et al., 2017; Chollet, 2017). To mitigate aliasing under temporal decimation, downsampling applies a fixed depthwise smoothing filter followed by strided averaging ("blur-pool" style) (Zhang, 2019). A dilated bottleneck further enlarges the receptive field without additional downsampling (Yu and Koltun, 2016).

Segmentation model zoo. Table 3 lists the variants used for ensembling/stacking. ETS-U-Net-L/M/S share the same ETS-U-Net template and vary width and the number of residual refinement blocks per level. Attention ETS-U-Net replaces the bottleneck with temporal multi-head self-attention (MHSA) (Vaswani et al., 2017) and uses stochastic depth (DropPath) regularization (Huang et al., 2016) to stabilize training at higher capacity. EEG-Inception Segmentation replaces single-path temporal filtering with Inception-style multi-branch temporal kernels (Szegedy et al., 2015), following the EEG-Inception design for ERP feature extraction (Santamaría-Vázquez et al., 2020). Finally, Factorized ETS-U-Net augments the channel-mixing front-end with factorized cross-channel interactions inspired by Factorization Machines (FM) (Rendle, 2010), optionally

Table 3: Segmentation model variants used for reaction-time prediction. Each model outputs per-time logits and is trained under the same segmentation objective; architectural diversity is exploited for ensembling/stacking.

Model	**Architecture**	**Key features**	**Strengths**
ETS-U-Net-L (wide)	ETS-U-Net	wider encoder; depthwise-separable convs; 5 Res-Blocks/level; dilated bottle-neck	highest capacity; captures complex local temporal patterns
ETS-U-Net-M	ETS-U-Net	moderate width; depthwise-separable convs; 5 Res-Blocks/level; dilated bottle-neck	best single-model accuracy / efficiency tradeoff
ETS-U-Net-S	ETS-U-Net	lighter encoder; depthwise-separable convs; 3 Res-Blocks/level; same bottle-neck family	fast, low-variance; comple-ments deeper models
Attention ETS-U-Net (MHSA)	Attention ETS-U-Net	MHSA bottleneck (4 heads, depth=3); DropPath; posi-tional depthwise conv	captures long-range tempo-ral dependencies
EEG-Inception Segmentation	EEG-Inception 1D	multi-branch temporal scales (0.5/0.25/0.125 s) with feature fusion	strong multi-scale feature extraction
Factorized ETS-U-Net	Factorized ETS-U-Net	FM-style cross-channel in-teractions at input; optional per-stage FM injection	distinct cross-channel in-ductive bias

also injected within encoder stages to introduce a distinct cross-channel inductive bias.

Architectural diversity is introduced to improve ensembling and stacking. ETS-U-Net-L (wide) uses a wider encoder and 5 residual blocks per level, providing the highest capacity. ETS-U-Net-M retains the same depth (5 blocks/level) with a more moderate width for a stronger accuracy–efficiency trade-off, while ETS-U-Net-S uses a lighter encoder (3 blocks/level) and yields faster, lower-variance base learners. Long-range temporal interactions are emphasized in Attention ETS-U-Net (MHSA) via a multi-head self-attention bottleneck. EEG-Inception Segmentation replaces single-path temporal filtering with multi-branch kernels for multi-scale feature extraction. Finally, Factorized ETS-U-Net augments the standard channel-mixing front-end with an FM-style factorized interaction module (optionally also injected within encoder stages), providing a distinct cross-channel inductive bias. The post-hoc distribution-aware stacking procedure used for the secondary ensemble analysis is described in Appendix A.

4.4 Soft-Label Distributional Supervision

Soft target distribution. For a sampled crop, the ground-truth event time within the crop is represented as a relative time $y_{\text{rel}} \in [0, \text{crop\_sec})$ with $\text{crop\_sec} = 2$. A smooth segmentation label is constructed using a Gaussian kernel with bandwidth σ:

$$\tilde{q}_t = \exp\left(-\frac{(g_t - y_{\mathrm{rel}})^2}{2\sigma^2}\right), \qquad q_t = \frac{\tilde{q}_t}{\sum_{k=0}^{T-1}\tilde{q}_k}, \qquad \sum_{t=0}^{T-1} q_t = 1. \tag{1}$$

This smoothing stabilizes optimization while preserving temporal localization; in practice, σ can be annealed (start larger and decrease over training).

Model output and losses. A 1-D segmentation backbone produces logits z_t for $t = 0,\dots,T-1$, and a predicted event-time distribution is obtained via temperature-scaled softmax,

$$p_t(\tau) = \frac{e^{z_t/\tau}}{\sum_{k=0}^{T-1} e^{z_k/\tau}}. \tag{2}$$

The soft-label segmentation family is written as a weighted objective:

$$\begin{aligned}\mathscr{L} = {} & \lambda_{\mathrm{time}}\,\mathrm{RMSE}\big(\hat{t}_{\mathrm{rel}}(\tau), y_{\mathrm{rel}}\big) + \lambda_{\mathrm{CE}}\,\mathrm{CE}\big(q, p(\tau)\big) \\ & + \lambda_{W1}\,W_1\big(\mathrm{CDF}(q), \mathrm{CDF}(p(\tau))\big) + \lambda_{\mathrm{KL}}\,D_{\mathrm{KL}}\big(q \| p(\tau)\big) - \lambda_H\,H\big(p(\tau)\big),\end{aligned} \tag{3}$$

where

$$\mathrm{CE}(q,p) = -\sum_{t=0}^{T-1} q_t \log p_t,\; D_{\mathrm{KL}}(q\|p) = \sum_{t=0}^{T-1} q_t \log\frac{q_t}{p_t},\; H(p) = -\sum_{t=0}^{T-1} p_t \log p_t.$$

This expression defines an ablation family rather than a single mandatory loss cocktail. CE-only supervision sets the scalar time term to zero and asks the model to match a Gaussian soft event-time label. The hybrid objective combines CE with a soft-argmax time loss. The time-only ablation removes distributional matching while keeping the same readout, and the Wasserstein-only ablation tests whether a CDF-based geometry loss is sufficient on its own. The entropy, KL, focal, and related terms are best viewed as posterior-geometry controls rather than as the main contribution. We also evaluate a hazard-parameterized readout as an extension, mapping logits to a discrete-time survival process before forming the event-time posterior.

Release-separated segmentation comparison. Table 4 summarizes the event-time segmentation loss ablations, with the likelihood-family extensions defined in Section 4.5 included for comparison. The calibrated hybrid event-time segmentation model reaches 0.935573 nRMSE on R11, improving over the strongest direct-regression baseline. CE-only soft-label supervision is also strong. The Gaussian-mixture and hazard rows are compact EventNLL-family extensions: the former changes the observation kernel, while the latter changes the event-time posterior parameterization. This is the central empirical step in the narrative: once RT is represented as an event-time posterior, distributional supervision over time becomes more useful than treating RT as a scalar window label.

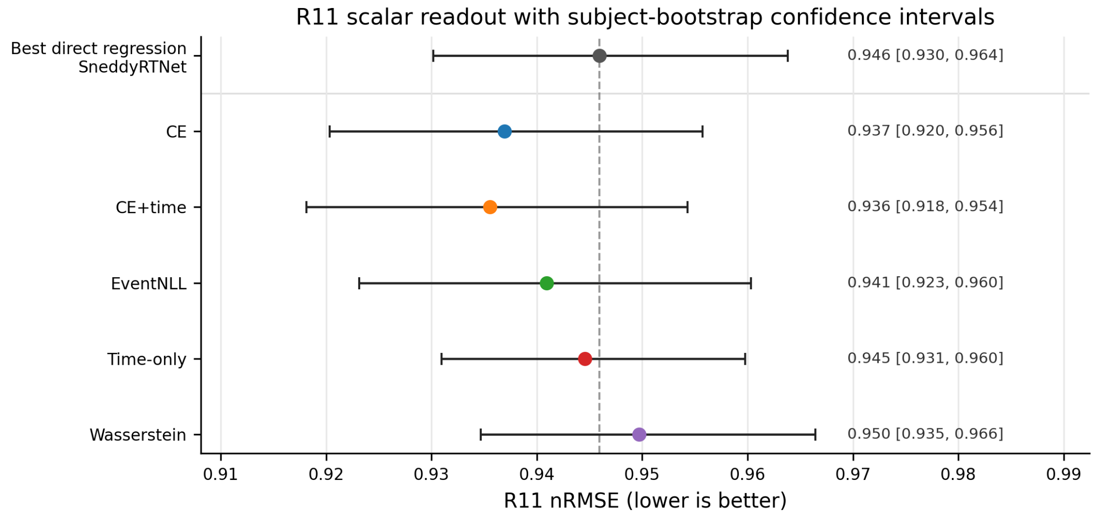

Figure 5: R11 scalar readout performance with subject-bootstrap confidence intervals. The dashed vertical line marks the strongest direct-regression baseline. Several event-time losses improve scalar nRMSE while remaining close enough to motivate posterior-level analysis rather than ranking losses only by point-prediction error.

Table 4: Completed event-time segmentation loss ablations under the release-separated protocol. "Cal." denotes post-hoc temperature selection on R9-R10 and one-shot application to R11.

Run	**Objective**	τ	**Valid nRMSE** $\downarrow$	**R11 nRMSE** $\downarrow$
CE only, base	soft-label event-time CE	0.65	0.922529	0.938738 [0.921696, 0.958015]
CE only, cal.	soft-label event-time CE	0.70	0.922277	0.936942 [0.920329, 0.955704]
Hybrid, base	soft-label CE + soft-argmax time loss	0.65	0.925601	0.940006 [0.921243, 0.960020]
Hybrid, cal.	soft-label CE + soft-argmax time loss	0.80	0.924544	0.935573 [0.918136, 0.954309]
Event NLL, cal.	continuous latent event-time likelihood	0.70	0.923619	0.940947 [0.923160, 0.960322]
Gaussian mixture NLL, cal.	two-scale EventNLL observation kernel	0.75	0.921676	0.939267 [0.921864, 0.958291]
Hazard EventNLL	hazard-parameterized event-time posterior	0.65	0.924318	0.937332 [0.920594, 0.955704]
Time only, cal.	soft-argmax scalar time loss only	0.85	0.935298	0.944578 [0.930935, 0.959775]
Wasserstein only, cal.	event-time CDF distance only	2.95	0.936941	0.949716 [0.934666, 0.966443]

Objective ablations. The loss ablations separate the value of the temporal readout from the value of distributional supervision. The time-only model directly optimizes soft-argmax error but removes distribution matching; it is weaker than CE-only and hybrid segmentation. The Wasserstein-only run is a useful negative control: not every distributional distance recovers the gain. CE, CE+time, and EventNLL remain in the same performance band, supporting the view that the main gain comes from the event-time formulation rather than from a single hand-tuned loss. The hazard EventNLL result further shows that the formulation is not tied to a categorical softmax readout: a survival-style posterior reaches the same scalar-performance regime while preserving the event-time likelihood view.

Table 5: Five-seed robustness summary for selected core models. Values are mean ± sample standard deviation over seeds 2025–2029; R11 range is min–max across seeds. Posterior scores are reported only for event-time segmentation runs. Lower is better for all metrics.

Model	Role	Valid nRMSE ↓	R11 nRMSE ↓	R11 range	Posterior CRPS ↓	Fixed NLL ↓
ETR-CNN	temporal-readout scalar baseline	0.9359 ± 0.0031	0.9495 ± 0.0048	0.9451–0.9570	–	–
TIDNet wrapped	strongest wrapped external baseline	0.9546 ± 0.0014	0.9614 ± 0.0046	0.9547–0.9656	–	–
ETS-U-Net CE	soft-label event-time CE	0.9209 ± 0.0024	0.9384 ± 0.0021	0.9352–0.9407	0.1927 ± 0.0007	0.3062 ± 0.0086
ETS-U-Net EventNLL	latent event-time likelihood	0.9242 ± 0.0024	0.9391 ± 0.0045	0.9318–0.9439	0.2011 ± 0.0019	0.1523 ± 0.0133
ETS-U-Net time-only	soft-argmax time loss control	0.9371 ± 0.0020	0.9473 ± 0.0021	0.9447–0.9496	0.2040 ± 0.0020	0.3878 ± 0.0268

Seed robustness. To test whether the main comparison depends on a favorable random initialization, we repeated selected core recipes over five seeds (2025–2029). This analysis is a reliability check rather than a new model-selection stage: it focuses on the strongest temporal-readout regression baseline, the strongest wrapped external baseline, the two main distributional event-time objectives, and the time-only segmentation control.

Table 5 shows that the central event-time supervision gap is stable. CE and EventNLL segmentation remain about 0.010–0.011 R11 nRMSE better than the repeated ETR-CNN scalar baseline on average. In contrast, the time-only segmentation control is much closer to ETR-CNN, supporting the interpretation that the gain comes from distributional event-time supervision rather than only from a U-Net backbone or soft-argmax readout. The optional repeated TIDNet baseline also remains weaker than both ETR-CNN and the event-time segmentation variants.

4.5 Latent Event-Time Likelihood

As a probabilistic alternative to soft Gaussian-label supervision, we also evaluate a latent event-time likelihood. The temporal softmax defines a posterior over latent event bins, and the observed scalar RT is modeled as a noisy realization of the latent event time:

$$p(y \mid X) = \sum_{t=0}^{T-1} p_t(X)\, \mathcal{N}\left(y; t_0 + g_t,\, \sigma_y^2\right), \qquad \mathcal{L}_{\text{EventNLL}} = -\log p(y \mid X). \tag{4}$$

This objective keeps the same event-time interpretation but replaces the artificial target mask q_t with a marginal likelihood over latent response-relevant event times. We therefore present EventNLL primarily as a more principled latent-variable objective, not as the scalar-NRMSE winner among the evaluated losses.

Observation-kernel and hazard extensions. The EventNLL formulation separates the latent event-time posterior $p_t(X)$ from the observation model $K(y \mid t)$. The default model uses a single Gaussian observation kernel, but we also evaluate a two-scale Gaussian mixture,

$$K(y \mid t) = (1-\alpha)\,\mathcal{N}(y; t_0 + g_t, \sigma^2_{\text{narrow}}) + \alpha\,\mathcal{N}(y; t_0 + g_t, \sigma^2_{\text{wide}}), \tag{5}$$

which keeps most probability mass in a narrow timing error while allowing a wider tail for occasional noisy RT observations. In the completed run, $\sigma_{\text{narrow}} = 0.10$ s, $\sigma_{\text{wide}} = 0.35$ s, and $\alpha = 0.10$.

As a readout extension, we also evaluate a hazard/survival parameterization. Instead of using logits only as unconstrained softmax scores, each temporal logit defines a conditional event probability

$$h_t = \sigma(z_t/\tau), \qquad p_t^{\text{haz}} \propto h_t \prod_{k<t}(1-h_k), \tag{6}$$

with the resulting event-bin probabilities normalized over the represented window. This keeps the same event-time likelihood in Eq. 4, but changes the posterior parameterization from a categorical softmax to a discrete-time survival process. We treat this as a paper-facing extension because it tests whether the event-time formulation is tied to one softmax implementation.

Likelihood-family comparison. The EventNLL rows in Table 4 show that the event-time NLL run nearly matches CE-only and hybrid segmentation without requiring a hand-weighted CE+time objective. We therefore present it as a principled latent-variable likelihood rather than as a claim of best scalar nRMSE. The Gaussian-mixture kernel improves the EventNLL-family scalar readout relative to the fixed Gaussian kernel, while the hazard EventNLL extension reaches CE-level R11 performance without reverting to scalar regression. The posterior-geometry analysis in Table 6 further shows that EventNLL produces a much narrower posterior than CE or CE+time and places more mass near the target, but it also under-covers nominal central intervals. This makes it scientifically useful: it reveals that similar scalar errors can correspond to different posterior geometries.

4.6 Posterior Geometry and Readout Temperature Tuning

The posterior view makes geometric properties of the predicted distribution directly inspectable. CDF-based distances penalize temporal displacement of probability mass, entropy and variance summarize posterior sharpness, and quantiles expose asymmetric uncertainty around the predicted response time. These summaries are useful for diagnosis and for downstream stacking.

When used, explicit posterior regularization can be written generically as

$$\mathscr{L} = \mathscr{L}_{\text{event}} + \lambda R(p(t \mid X)), \tag{7}$$

where R may target entropy, variance, tail mass, or other posterior-shape statistics. In this manuscript we treat such terms as analysis and control of posterior geometry unless a clean ablation has been completed.

Inference and temperature tuning. The relative event time is estimated by expectation (soft-argmax),

$$\hat{t}_{\mathrm{rel}}(\tau) = \sum_{t=0}^{T-1} p_t(\tau)\, g_t, \qquad \hat{t}_{\mathrm{abs}}(\tau) = t_0 + \hat{t}_{\mathrm{rel}}(\tau), \tag{8}$$

where $g_t = t\,\Delta t$ denotes the discrete time grid. Under squared error, the Bayes-optimal point estimate is the conditional mean; accordingly, reading out RT by $\mathbb{E}_{p(t)}[g]$ aligns naturally with the nRMSE objective (RMSE up to a constant normalization).

In practice, different training choices (loss weighting, augmentation strength, and checkpoint selection) produced time-distributions with markedly different sharpness and entropy: some models generated very peaked $p(t)$ while others were systematically more diffuse. This creates a confidence-scale mismatch, where naive averaging (or stacking without temperature-scaled inputs) can underweight accurate but underconfident models and overweight overconfident ones.

We therefore use post-hoc temperature scaling (Guo et al., 2017) to tune the softness of $p(t)$, improving the readout and making confidence profiles more comparable across losses and models. Temperature is selected by validation tuning for each base model m:

$$\tau_m^\star = \arg\min_{\tau \in \mathscr{T}_{\mathrm{cal}}} \ \mathrm{nRMSE}_{\mathrm{val}}\big(\hat{t}_{\mathrm{abs}}^{(m)}(\tau)\big). \tag{9}$$

Under the release-separated protocol, $\mathscr{T}_{\mathrm{cal}}$ is selected on the R9-R10 development split and applied unchanged to R11.

Because confidence scales differ across objectives and architectures, temperature is treated as a posterior-geometry control selected on development data rather than as a separate model-selection signal.

Posterior geometry and interval coverage. The scalar comparisons above compare models through the posterior mean, but the event-time formulation produces a full distribution over response latencies. Two losses can therefore have similar nRMSE while assigning probability mass very differently over time. We analyze this posterior geometry to ask whether the scalar prediction is supported by a localized temporal event distribution, whether the posterior is overly diffuse or overly concentrated, and whether nominal posterior intervals have meaningful empirical coverage.

For this analysis, the softmax temperature is selected on R9–R10 to minimize scalar nRMSE of the posterior-mean readout. Thus, we treat temperature selection as readout-temperature tuning rather than as full probabilistic calibration. After this readout tuning, we evaluate the learned event-time posterior itself: whether its mass is near the observed RT, whether the posterior mean agrees with the posterior mode, and whether nominal posterior intervals contain the true RT at the expected rate.

Figure 7 provides a trial-sorted view of these posteriors before reducing them to summary metrics. It shows how each loss distributes probability mass across the represented RT range, making posterior width, target alignment, and loss-specific confidence patterns visible at the dataset level.

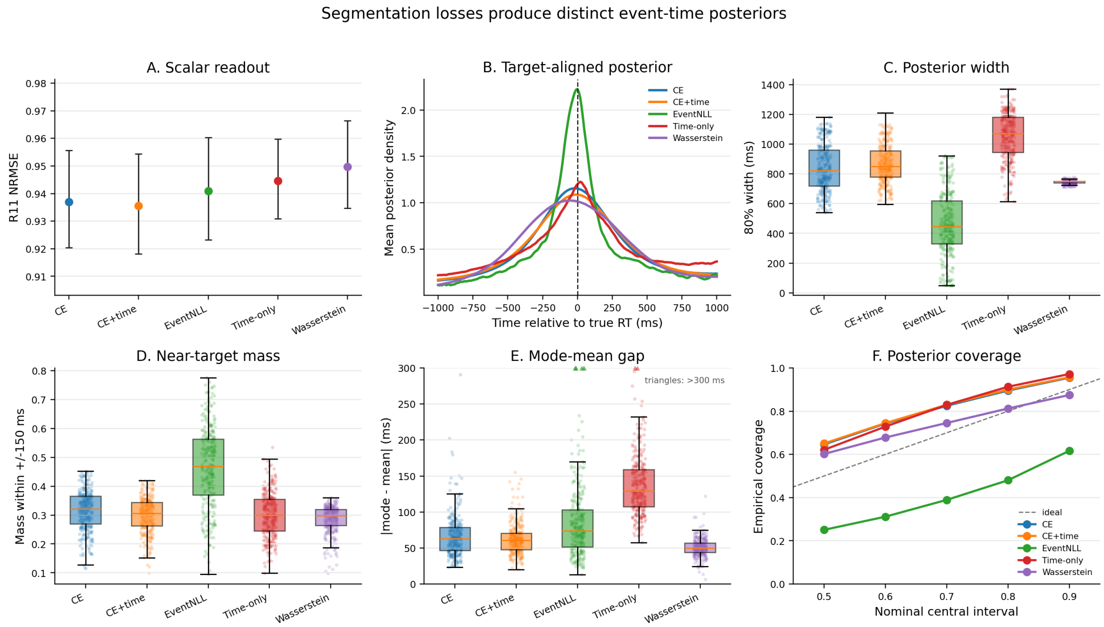

Figure 6: Posterior geometry differs across event-time losses even when scalar nRMSE is similar. Panel A reports R11 scalar readout after selecting the softmax temperature on R9–R10 to minimize nRMSE. Panel B aligns predicted posteriors to the true RT. Panels C–E summarize posterior width, near-target mass, and mode-mean disagreement. Panel F compares empirical coverage of central posterior intervals; the diagonal is ideal interval coverage.

Table 6 uses four groups of metrics. First, scalar nRMSE and MAE measure the posterior-mean RT prediction. Second, CRPS and fixed-kernel EventNLL score the full posterior distribution; lower values indicate better distributional predictions. The fixed-kernel EventNLL uses the same Gaussian observation kernel for all models, so it evaluates the predicted event-time distribution under a common likelihood score rather than under each model's own training loss. Third, Width80, Mass ± 150 ms, and mode-mean gap describe posterior geometry: Width80 is the median width of the nominal 80% central posterior interval, Mass ± 150 ms is the posterior probability assigned within 150 ms of the observed RT, and mode-mean gap measures disagreement between the posterior peak and the posterior mean used for scalar readout. Finally, Coverage80 and Coverage MAE summarize interval coverage. Coverage80 is the empirical coverage of the nominal 80% central interval, whose calibrated target is 0.80. Coverage MAE is the mean absolute difference between empirical and nominal coverage over central intervals $\{50\%, 60\%, 70\%, 80\%, 90\%\}$.

These metrics show that the losses induce different output semantics. CE+time and CE give the strongest scalar readouts, but their nominal 80% intervals cover about 90% of trials, indicating conservative over-coverage. EventNLL is not well calibrated as an uncertainty posterior: its nominal 80% interval covers only 48% of trials. Its value is instead localization-like concentration, with the narrowest Width80, the highest near-target mass, and the best fixed-kernel EventNLL score. Wasserstein has the smallest

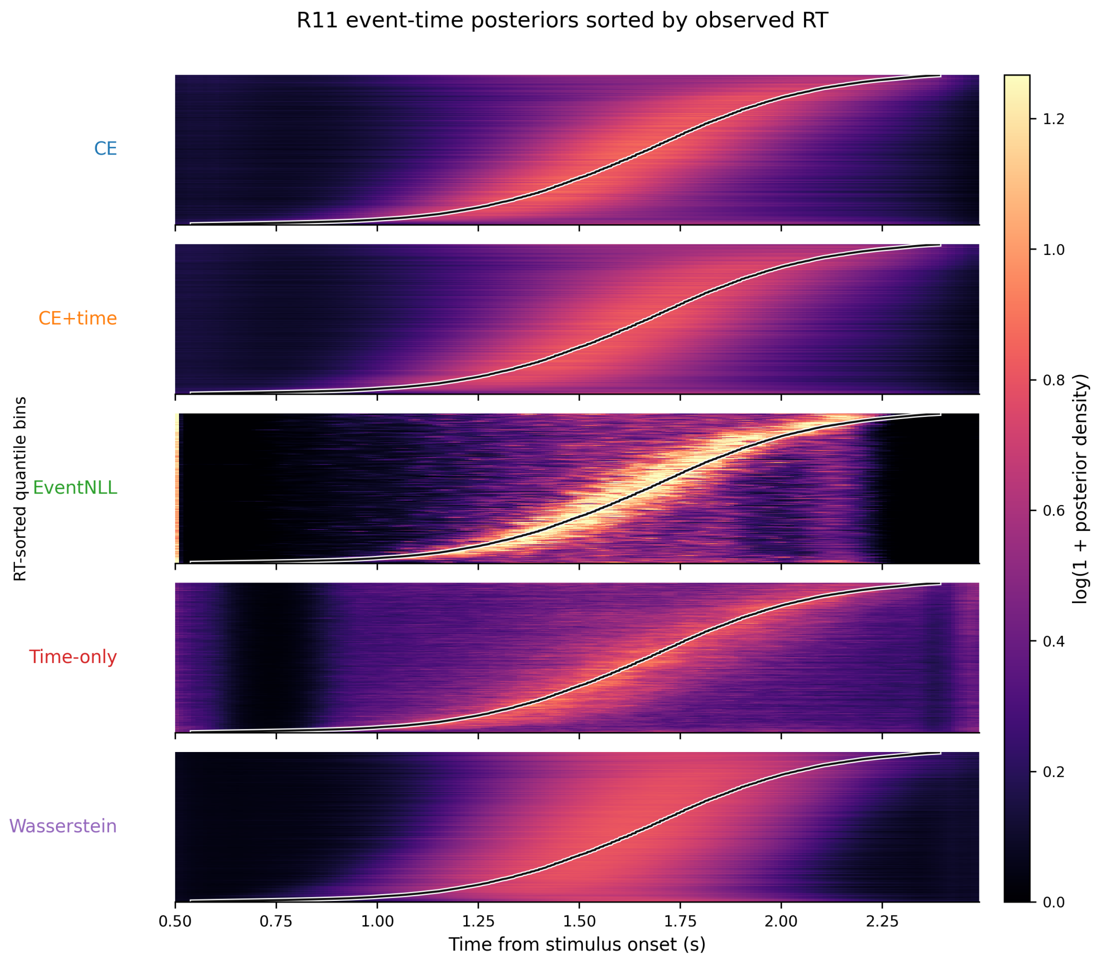

Figure 7: Trial-sorted event-time posterior maps on R11 for matched segmentation losses. Rows are quantile bins of representable R11 trials sorted by observed RT; the x-axis is time from stimulus onset; color shows log-transformed posterior density averaged within each bin. The black curve marks the mean observed RT in each bin.

Table 6: Quantitative posterior geometry on R11 for matched event-time segmentation losses. Metrics are defined in the text. CRPS is reported in milliseconds. Fixed-kernel EventNLL uses a common Gaussian observation kernel for all models ($\sigma = 0.15$ s). For Coverage80, values closer to 0.80 indicate better nominal 80% interval coverage; for Coverage MAE, lower is better.

Model	nRMSE	MAE ms	CRPS ms	Fixed NLL	Width80 ms	Mass $\pm$150 ms	Mode-mean gap ms	Coverage80	Coverage MAE
CE	0.937	216	155	0.115	820	0.328	57	0.895	0.113
CE+time	0.936	219	157	0.144	860	0.311	57	0.899	0.116
EventNLL	0.941	216	162	-0.027	450	0.487	72	0.480	0.290
Time-only	0.945	228	167	0.203	1070	0.311	129	0.913	0.113
Wasserstein	0.950	222	165	0.219	740	0.296	50	0.812	0.053

Coverage MAE and a Coverage80 value closest to 0.80, but this comes with weaker scalar nRMSE and lower near-target mass. Thus, posterior geometry reveals a trade-off among scalar accuracy, target concentration, and interval coverage that is hidden by nRMSE alone.

Secondary ensemble use of posterior features. The same posterior summaries can also serve as post-hoc ensemble features, but we treat this as a secondary analysis rather than as a main release-separated claim. The single-model R11 comparisons above are intentionally reported without stacking, so that the main evidence isolates the event-time objective, scalar readout, posterior geometry, and the shifted-crop diagnostic below. The stacker construction is described in Appendix A; in the original hidden-release CodaBench evaluation (Appendix Table 8), gradient boosting stacking achieved 0.91394 hidden nRMSE, compared with 0.92334 for the best single model, 0.91840 for manual blending, and 0.91539 for Ridge stacking. This supports posterior-derived meta-features as an ensemble calibration layer while keeping the paper's main claims tied to single-model release-separated analyses.

4.7 Shifted-Crop Localization Diagnostic

The posterior analyses above show that event-time losses produce different probability geometries, but they do not by themselves prove that a model has learned a within-crop event localizer. A fixed stimulus-locked window can still permit shortcut behavior: a model may predict trial difficulty, subject-level slowness, or stimulus-locked response tendency without moving its prediction when the crop is shifted. We therefore introduced a shifted-crop diagnostic on R11. For each trial, we used a 5 s EEG window and evaluated the same trained model on 2 s crops starting at $s \in \{0.2, 0.3, \ldots, 0.8\}$ s. The reference crop is $s = 0.5$ s, matching the training and main inference protocol. To avoid target-censoring effects, this analysis uses the common-inside subset where RT remains inside all shifted crops ($0.8 \leq \text{RT} \leq 2.2$ s; 14,183 trials).

If the model truly localizes the response-relevant event within the crop, shifting the crop start by Δs should move the predicted crop-relative event time by approximately $-\Delta s$, giving a raw shift slope near -1. A crop-invariant scalar regressor should instead have a slope near 0. Table 7 shows that the event-time segmentation models preserve or improve reference-crop accuracy and move in a more localizer-like direction than the best scalar regressor. EventNLL reaches a raw shift slope of -0.292 [95% CI: -0.311, -0.272], compared with -0.253 [-0.270, -0.235] for ETR-CNN and -0.172 on average across regression baselines. The fraction of localizer-like trials also increases from 10.1% for ETR-CNN to 14.8% for CE and 15.9% for EventNLL.

This result should be interpreted as evidence for a representational shift, not as solved shift equivariance. The segmentation models remain far from the ideal slope of -1, indicating that fixed-window training still mixes event localization with scalar response-tendency shortcuts. That limitation is useful: it identifies shift-augmented, crop-relative event-time training as the natural next step if the goal is a stronger trial-wise neural event localizer.

Table 7: Shifted-crop diagnostic on the R11 common-inside subset. The ideal within-crop event localizer has raw shift slope -1; a crop-invariant scalar shortcut has slope 0. Reference nRMSE is computed at the standard $s = 0.5$ s crop.

Model	Ref nRMSE ↓	Ref MAE, s ↓	Worst Δ nRMSE	Raw shift slope	Localizer-like trials	Invariant-like trials
ETR-CNN	0.874	0.206 [0.199, 0.215]	0.174	−0.253 [−0.270, −0.235]	10.1% [9.0, 11.1]	34.5%
ETS-U-Net CE	0.856	0.201 [0.193, 0.209]	0.193	−0.287 [−0.306, −0.267]	14.8% [13.3, 16.3]	34.4%
ETS-U-Net EventNLL	0.859	0.201 [0.193, 0.210]	0.198	−0.292 [−0.311, −0.272]	15.9% [14.5, 17.3]	33.3%

5 Discussion

5.1 RT Prediction as Event-Time Localization

The results support treating trial-wise RT prediction as event-time localization rather than as ordinary scalar regression. The label is the latency of a response within a post-stimulus interval, and models benefit when that timing structure is explicit in both the output space and the loss. The shifted-crop diagnostic adds a stricter check on this interpretation: event-time segmentation models move in a more event-like direction than scalar regression under temporal crop shifts, but their slopes remain far from perfect localizer behavior. This framing therefore connects naturally to response-locked EEG analyses while also making clear that fixed-window RT prediction is not the same as fully shift-equivariant event detection.

5.2 Why Distributional Supervision Helps

Distributional supervision gives the model a local temporal target instead of a single global scalar penalty. CE-only and hybrid losses outperform the time-only loss, suggesting that the gain is not just a differentiable soft-argmax readout. The soft event-time target encourages probability mass near plausible response latencies and makes label-consistent temporal crop jitter easier to justify. EventNLL reaches a similar scalar-error regime through a different route: it marginalizes a latent event time instead of asking the model to imitate a hand-constructed Gaussian mask.

5.3 Relation to EEG Backbones and Pretraining

The wrapped external baselines show that standard EEG architectures can train under the fixed normalization protocol, but they do not close the gap to compact task-specific models or event-time segmentation. Foundation-style architectures are evaluated here as architectures trained from scratch, not as pretrained models. The comparison is therefore not a test of pretraining; it shows that, under this benchmark protocol, formulation and objective design are strong levers even with lightweight supervised models.

5.4 Why Posterior Geometry Matters

The event-time posterior contains more information than its expectation. Sharpness, entropy, variance, quantiles, and the gap between posterior mode and posterior mean describe confidence and ambiguity in ways that scalar predictions discard. The posterior analysis shows that CE, CE+time, EventNLL, time-only, and Wasserstein losses can have similar scalar readouts but different target-alignment, width, and coverage behavior. Temperature calibration is therefore best understood as posterior-geometry control and a fairness step for comparing readouts, not as the main methodological claim.

5.5 Limitations

The main limitation is that all claims are tied to HBN-EEG CCD and the release-separated protocol used here. R11 is a stronger local holdout than reused validation performance, but it is still drawn from the same broader dataset family and preprocessing pipeline. The shifted-crop diagnostic shows partial localization, not full shift equivariance, so stronger claims about trial-wise neural event localization would require training protocols that explicitly randomize crop position and supervise crop-relative timing. The five-seed robustness checks support the main scalar-versus-event-time comparison, but they cover selected core recipes rather than every kernel, hazard, calibration, and architecture variant. The foundation-style comparisons also do not use pretrained weights or montage-specialized setup changes; they are controlled architecture baselines rather than a full evaluation of EEG foundation models.

6 Conclusion

We reformulated EEG-based reaction-time prediction as event-time distribution modeling and evaluated it under a release-separated HBN-EEG protocol. The results indicate that supervising an event-time posterior improves RT prediction beyond scalar regression and beyond a temporal-readout head alone. Loss ablations and five-seed robustness checks show that distributional event-time supervision matters, EventNLL provides a principled latent-variable likelihood alternative, and posterior-geometry plus shifted-crop analyses reveal output behavior that scalar nRMSE alone cannot capture.

A Distribution-Aware Stacking

Segmentation-style outputs enable richer ensembling than scalar averaging, but combination becomes sensitive to model confidence: naive averaging can overweight overconfident models and underweight accurate but underconfident ones. We therefore define an auxiliary second-stage stacker on distributional meta-features extracted from the time-distribution outputs.

In a release-separated application, base segmentation models are fit on R1-R8 before the stacker is trained. Meta-model inputs are then computed on the R9-R10 development partition using subject-disjoint folds, so that the stacker never receives in-fold predictions for the same subject. Logits and temperature-scaled event-time probabilities are converted to meta-features such as peak time, entropy, variance, expected time, and CDF

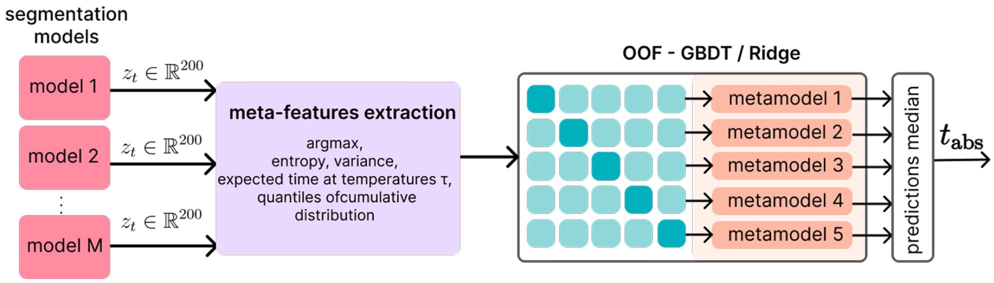

Figure 8: OOF stacking for RT prediction from time-distribution outputs. Per-model logits/distributions are converted into meta-features (e.g., peak time, entropy/variance, temperature-conditioned expected times, and CDF quantiles) and combined by a second-stage stacker (Ridge/GBDT). Predictions from fold-trained stackers are aggregated at inference (median) to produce the final RT estimate.

quantiles. This stacker is treated as a post-hoc ensemble layer rather than as part of the single-model event-time comparison.

Meta-features from logits and temperature-scaled probabilities. Meta-features are extracted per model from (i) raw logits and (ii) temperature-scaled probabilities computed for multiple temperatures $\tau \in \mathscr{T}_{\text{meta}}$. From logits, a hard-peak estimate and confidence proxies are computed:

$$t^\star = \arg\max_t z_t, \qquad \hat{t}_{\text{hard}} = t_0 + g_{t^\star}, \qquad z_{\max} = \max_t z_t, \qquad m_{\text{top2}} = z_{\max} - \max_{t \neq t^\star} z_t. \tag{10}$$

For each $\tau \in \mathscr{T}_{\text{meta}}$, probabilities are formed as $p(\tau) = \text{softmax}(\mathbf{z}/\tau)$ (Eq. 2, Sec. 4.3), where $\mathbf{z} = (z_0, \ldots, z_{T-1})$ denotes the per-time logit vector for a given model and window. Distributional summaries are computed, including the soft-argmax time (Eq. 8, Sec. 4.3), entropy $H(p(\tau))$, peak probability $p_{\max}(\tau)$, time variance $\text{Var}_{p(\tau)}[g]$, and CDF time-quantiles (10/50/90%). In our implementation, $\mathscr{T}_{\text{meta}} = \{0.7, 1.0, 1.3\}$.

Meta-learner and inference. We use linear Ridge regression as a strong baseline meta-learner, and gradient boosting to capture non-linear reliability patterns across models and temperatures. The meta-learner is trained using subject-disjoint cross-validation: each subject is assigned to exactly one fold (Figure 8). For fold k, the stacker is fit on the other folds and predicts RT for the held-out fold. The leakage-controlled diagnostic is the out-of-fold nRMSE from these held-out predictions. Folds are stratified to match the per-subject mean RT distribution across folds. For held-out inference, the same feature construction is applied to unseen logits; predictions from the fold-trained stackers are aggregated to obtain the final RT estimate. Manual blending serves as a reference, but it cannot adapt weights to sample-specific confidence profiles.

B Original Hidden CodaBench Benchmark Results

We report the original hidden-release CodaBench evaluation as historical challenge context. These scores provide continuity with the original benchmark setting, while the main empirical conclusions in this manuscript are based on the labeled R11 release with subject-bootstrap confidence intervals.

Table 8: Original hidden-release CodaBench RT results from the earlier submission-oriented pipeline. These results are reported for historical comparison with the challenge setting.

Method	**Validation nRMSE** $\downarrow$	**Hidden nRMSE** $\downarrow$
Best temperature-tuned single model	0.898757	0.92334
Manual blending	0.899979	0.91840
Ridge stacking	0.890430	0.91539
Gradient boosting stacking	**0.883365**	**0.91394**

C Unwrapped External Backbone Sanity Checks

Unwrapped external architectures are useful for diagnosing scale sensitivity. Because our own models use per-window standardization, the primary comparison uses wrapped external models. Table 9 reports matched sanity checks for representative off-the-shelf architectures with and without the fixed per-window standardization wrapper. The unwrapped rows often collapse toward target-standard-deviation behavior (nRMSE near 1), whereas the wrapped variants train meaningfully under the same release-separated protocol used in the main tables.

D Additional Protocol Calibration Runs

Protocol calibration evaluated optimizer choice, learning rate, soft-label width, batch size, and augmentation strength before fixing the training recipe used in the release-separated experiments. These diagnostics informed the protocol but are separated from the R11 results used for the main claims.

E Additional EventNLL Kernel and Hazard Robustness Checks

The main text keeps the event-time segmentation table focused on the core CE, EventNLL, and control objectives, plus two compact EventNLL-family extensions. Table 10 reports the broader kernel and hazard robustness checks. These runs use

Table 9: Unwrapped external-backbone sanity checks. "Wrapped" means that the model receives the same per-window channel standardization used by our compact baselines. Lower nRMSE is better. These checks diagnose scale sensitivity; the wrapped variants are the fair external comparisons used in the main results.

Architecture	Unwrapped valid	Unwrapped R11	Wrapped valid	Wrapped R11	Takeaway
EEGNet	1.000000	1.002046 [0.999718, 1.007253]	0.957877	0.965041 [0.954684, 0.977111]	Standardization makes the compact EEG baseline train
EEGConformer	1.000000	1.000413 [0.999906, 1.003179]	0.957968	0.962802 [0.951634, 0.975382]	Transformer-style baseline is scale-sensitive without wrapping
LaBraM	1.000000	1.000711 [0.998638, 1.004512]	0.958254	0.965382 [0.952576, 0.979330]	From-scratch foundation-style architecture needs the fixed normalization

Table 10: Additional EventNLL kernel and hazard robustness checks. Lower nRMSE is better. "Cal." denotes post-hoc temperature selection on R9–R10; the heteroscedastic run is reported at its base readout because it was not part of the calibrated kernel table.

Variant	Change tested	τ	Valid nRMSE $\downarrow$	R11 nRMSE $\downarrow$	Takeaway
Gaussian EventNLL, cal.	fixed Gaussian observation kernel	0.70	0.923619	0.940947 [0.923160, 0.960322]	Core latent event-time likelihood reference
Gaussian mixture EventNLL, cal.	narrow/wide Gaussian observation mixture	0.75	0.921676	0.939267 [0.921864, 0.958291]	Strongest calibrated EventNLL-family scalar readout
Laplace EventNLL, cal.	Laplace observation kernel	0.75	0.926636	0.940696 [0.923718, 0.959285]	Robust kernel did not improve over cleaner Gaussian/mixture variants
Student-t EventNLL, cal.	heavy-tailed Student-t observation kernel	0.80	0.923259	0.941127 [0.923382, 0.960687]	Calibration recovers validation gap, but R11 remains tied or weaker
Heteroscedastic EventNLL	trial-wise learned observation scale $\sigma(x)$	0.65	0.923056	0.942343 [0.923630, 0.962703]	Learned scale did not improve R11 enough for the core table
Hazard EventNLL	hazard posterior with continuous Gaussian EventNLL	0.65	0.924318	0.937332 [0.920594, 0.955704]	Strong alternative posterior parameterization
Hazard discrete NLL	exact-bin discrete survival likelihood	0.65	0.926288	0.946823 [0.928204, 0.967220]	Negative control: exact-bin survival loss is weaker than continuous EventNLL

the same release-separated protocol and test whether the probabilistic event-time result depends on one specific Gaussian observation kernel or one specific softmax parameterization.

Overall, these checks support two points. First, the EventNLL view is not tied to a single Gaussian-softmax implementation: the hazard EventNLL and Gaussian-mixture kernel remain competitive. Second, more robust or more flexible observation models are not automatically better; Laplace, Student-t, heteroscedastic, and exact-bin hazard variants are useful robustness probes but do not strengthen the core R11 story enough to feature in the main table.

F Additional Architecture Ablations

Architecture ablations separate formulation effects from capacity and backbone-specific effects. In particular, ETS-U-Net capacity variants, attention bottlenecks, factorized channel interactions, recurrent variants, and EEG-Inception-style segmentation provide complementary checks on whether the event-time formulation depends on a single encoder-decoder instantiation.

G Additional Posterior Geometry Examples

Table 6 reports the quantitative posterior metrics behind Figure 6, and Figure 7 shows the corresponding trial-sorted posterior maps in the main text. Figure 9 shows representative trials where models have similar scalar predictions but different posterior widths, large mode-mean gaps, or strong disagreement across losses, and Figure 10 shows the development-split temperature curves used to choose post-hoc readout temperatures.

Figure 9: Representative calibrated event-time posteriors. Dashed black lines denote the true RT, and dotted colored lines denote model-specific posterior means. The examples illustrate why scalar RT alone is insufficient: models can agree on the mean while assigning very different posterior width, or disagree strongly because the posterior is multimodal or skewed.

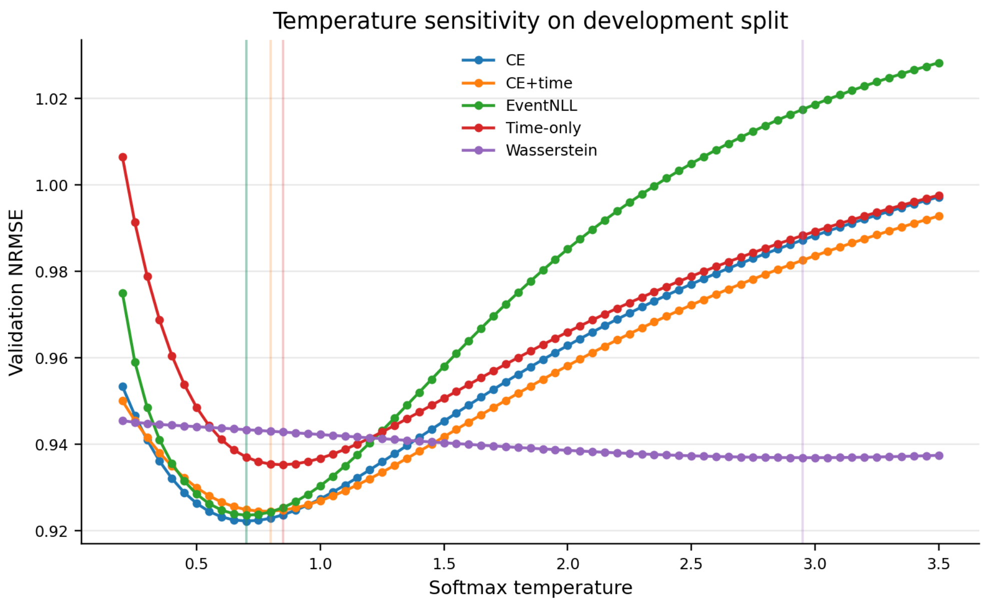

Figure 10: Temperature sensitivity on the R9-R10 development split. Vertical lines mark selected temperatures. The curves show that different losses induce different confidence scales, motivating temperature calibration as posterior-geometry control before comparing scalar readouts or using posterior features downstream.

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